Oriented Bounding Box Representation Based on Continuous Encoding in Oriented SAR Ship Detection

Peng Li, Cunqian Feng, Weike Feng, and Xiaowei Hu

***Abstract*—Ship detection of synthetic aperture radar (SAR) images holds significant value for both civilian and military applications. Compared to horizontal ship detection, oriented ship detection based on oriented bounding boxes can capture the orientation and aspect ratio of ships, thus receiving increasing attention. However, in oriented ship detection, when ships rotate near the specific angles, the angle prediction result obtained by the deep learning network may have a severe mutation, which is the well-known boundary discontinuity problem. To address the issue of boundary discontinuity in SAR ship detection, researchers have proposed numerous methods. However, through our systematic analysis, we found that these methods do not fundamentally solve the problem. To this end, we first clarified the reasons for the existence of boundary discontinuity and how it affects the detection network. Based on this, we proposed the conditions that the encoding methods and loss functions of the detection network must satisfy to address the issue of boundary discontinuity. In line with these conditions, we designed a continuous encoding method called coordinate decomposition method (CDM). In addition, we also analyzed the impact of different optimization methods on the detection network and, based on this, presented a joint optimization paradigm based on continuous encoding. Experimental results on two commonly used SAR ship detection datasets demonstrate that our proposed CDM encoding method effectively addresses the boundary discontinuity issue and enhances the detection performance. Compared to the state-of-the-art methods, the fully convolutional one-stage network using the CDM-based joint optimization achieves optimal detection results without employing any additional techniques.**

***Index Terms*—Boundary discontinuity, joint optimization, oriented object detection, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a microwave imaging sensor that works all day and is very little affected by weather conditions [1]. As a result, SAR systems have unique advantages in applications, such as ocean monitoring [2], military reconnaissance [3], and disaster warning [4]. Ship detection based on deep learning (DL) models is an important aspect of SAR maritime applications, which plays a significant role in coastal waterway monitoring, port management, and the safeguarding of coastal maritime security [5].

Conventional DL-based SAR ship detection models typically use horizontal bounding boxes (HBBs) to represent the position and size of ships [6], [7]. The loss function of the models is generally expressed by the intersection over union (IoU) of the ground truth (GT) HBB and the predicted HBB. While HBBs can effectively enclose the ship area, they fail to capture the ship's orientation and aspect ratio. Moreover, HBBs inevitably introduce a significant amount of background clutter, which is not conducive to subsequent SAR image processing [8].

To address this challenge, researchers have recently shifted their focus to oriented SAR ship detection algorithms that employ oriented bounding boxes (OBBs) [9], [10], [11], [12]. Unlike HBBs, OBBs are rectangular boxes with a rotation angle that can closely conform to the contours of ship. By using OBBs, it is possible to minimize background clutter and generate more accurate representations of ship aspect ratios and orientations.

Various DL-based detection models have been proposed for oriented SAR ship detection, in which OBBs are typically encoded as (cx, cy, w, h, θ), where (cx, cy) indicates the center position, (w, h) represents the width and height, and θ denotes the rotation angle. Since the Skew IoU (SkewIoU) between OBBs is difficult to calculate and nondifferentiable, the L1-based loss functions are generally selected as the regression loss for these models. However, DL models based on OBBs generally exhibit boundary discontinuity issues. The boundary discontinuity problem in this article does not refer to the boundary problem of the ship, but the boundary discontinuity in the angle parameter of the OBB. That is, when ship targets rotate near the boundary angle, the angle prediction experience sudden changes due to the angle periodicity, which is often called the boundary discontinuity problem. Boundary discontinuity leads to two issues: 1) when fitting discontinuous functions, DL models may have numerical instability, leading to erroneous angle predictions and 2) the inconsistencies between the L1-based loss function and the evaluation metrics make the DL model difficult to converge and train effectively. Hence, OBB encoding and loss function design are two typical methods for addressing the boundary discontinuity issue in oriented SAR ship detection.

In the context of oriented ship target detection, researchers have proposed many new encoding methods to represent OBBs. For example, Qiao et al. [13] proposed to represent the OBB

Received 15 January 2025; accepted 8 February 2025. Date of publication 13 February 2025; date of current version 3 March 2025. This work was supported by The Youth Innovation Team of Shaanxi Universities, the Youth Talent Lifting Project of the China Association for Science and Technology under Grant 2021-JCJQ-QT-018. *(Corresponding authors: Weike Feng; Xiaowei Hu.)*

The authors are with the Air and Missile Defense College, Air Force Engineering University, Xi'an 710051, China (e-mail: lipeng980404@163.com; fengcunqian@sina.com; fengweike007@163.com; xwhu002@163.com).

Digital Object Identifier 10.1109/JSTARS.2025.3541217

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

using the parameters of the center point of its enclosing rectangle, the length parameters, and the horizontal distance parameters from the center of the enclosing rectangle to two vertices of the OBB. Guo et al. [14] represented the angle using the distances from two vertices of the OBB to one of the upper-left corner vertices of its bounding rectangle. Ju et al. [15] encoded the OBB using a rectangle center, a mapped polar radius, and two polar angles. Although these methods avoid the boundary discontinuity problem caused by the angle periodicity, discontinuity may still occur during the process of encoding the OBB as a regression target, which is known as encoding discontinuity [16].

Average precision (AP) is an important evaluation metric for DL models, primarily dominated by SkewIoU for oriented ship detection. Due to the nondifferentiability and computational complexity of SkewIoU, it cannot be directly applied to oriented ship detection. Choosing the L1-based loss as the loss function for the DL model is a common practice. However, there is an inconsistency between the L1-based loss value and the evaluation metric [17]. In response to this problem, researchers have proposed some new loss functions. For example, Xu et al. [18] proposed a triangle distance IoU loss to approximate SkewIoU loss. In [19], the OBB was replaced by its inscribed ellipse, and the Gaussian Wasserstein distance (GWD) between the ellipses is used to replace the SkewIoU. However, while these methods alleviate the inconsistency between the loss function and the metric, they do not address the issues of angle and encoding discontinuity effectively.

In summary, there is currently no clear model design paradigm to systematically address boundary discontinuity in oriented SAR ship detection. To this end, this article analyzes theoretically the fundamental reasons for the occurrence of boundary discontinuity problems. Based on this theoretical analysis, we first propose the basic conditions that must be satisfied to solve the boundary discontinuity problem. Then, we introduce a continuous encoding method called CDM, which is based on coordinate decomposition and merely adds one bit to the encoding length. Based on CDM, we propose a new joint optimization paradigm that integrates continuous encoding with joint optimization loss to further improve the detection performance of DL networks. To validate the effectiveness of our approach, extensive experiments were conducted on two commonly used directional SAR ship detection datasets. The experimental results indicate that CDM effectively addresses the boundary discontinuity problem and enhances the DL model's detection accuracy. Furthermore, the proposed new joint optimization paradigm achieved optimal detection results compared with other state-of-the-art models.

The contributions of this article are as follows.

1) We conduct a systematic summary and analysis of existing oriented SAR ship detection models, clarifying the logic for addressing the boundary discontinuity issue, and found that these models do not completely resolve this problem.
2) We propose the fundamental principles that need to be met to address the issue of boundary discontinuity, and based on this, we introduce an angle-continuous encoding method called CDM.
3) We present a joint optimization paradigm that incorporates angle continuous encoding. The detection model based on this paradigm achieved competitive advantages in detection accuracy compared to other state-of-the-art detection models.

II. Related Work

A. Oriented SAR Ship Detection

As a branch of horizontal SAR ship detection, oriented SAR ship detection has been widely studied. In horizontal detection, the HBB parameters are represented using four values (cx, cy, w, h), where (cx, cy) indicates the center position and (w, h) represents the width and height. The box regression typically employs an IoU-based loss, which is a commonly used approach that does not have issues of boundary discontinuity. In oriented ship detection, the minimal enclosing OBB (cx, cy, w, h, θ) is widely used to represent oriented objects, where θ is the rotated angle of box [9], [20], [21]. Due to the computational complexity and nondifferentiability of SkewIoU, it was initially considered that the IoU-based loss cannot be used for oriented detection. As a result, early oriented detection methods typically use the L1-based loss for parameters (cx, cy, w, h, θ), optimizing each parameter individually [22], which is called independent optimization.

B. Research Motivation

Due to the angle periodicity, there is a common phenomenon of boundary discontinuity in oriented object detection. In Fig. 1(a) and (b), we explain this in the long edge 90 (LE90) and OpenCV (OC) definitions respectively [23].

In Fig. 1(a), we assume that there is a blue predicted OBB (0, 0, 60, 8, 88°) and a red GT OBB (0, 0, 60, 8, −90°) at the boundary position. The black dashed line indicates that only a very small angle rotation is needed to regress to the GT OBB, and the SkewIoU at this point is approximately 1. However, due to the periodicity of angle (PoA), the L1-based loss value is quite large at this time, which is inconsistent with the value of SkewIoU. This is because the angle has a sudden change after it exceeds the defined range, which makes the L1 loss function suddenly increase, resulting in the inconsistency between the evaluation metric and the loss value. For the LE90 definition method in Fig. 1(a), the direction of the blue dashed line is the direction of the decrease of the angle parameter, and the decrease of the angle parameter will lead to the decrease of the L1 loss function. Similarly, in the OC definition in Fig. 1(b), the direction of the blue dashed line is also the direction of the loss function decline. Since the predicted OBB will regress in the direction that reduces the loss, the actual regression direction will be the direction of the blue dashed line shown in Fig. 1. This actual regression direction will cause the SkewIoU between the predicted OBB and GT OBB to become smaller, thereby degrading the detection accuracy of the model.

A similar situation also occurs for the OC definition, as shown in Fig. 1(b). However, the OC definition method not only has the issue of PoA but also faces the problem of exchangeability of edges.

In order to avoid the problems caused by angle periodicity, some new encoding methods have been proposed to represent

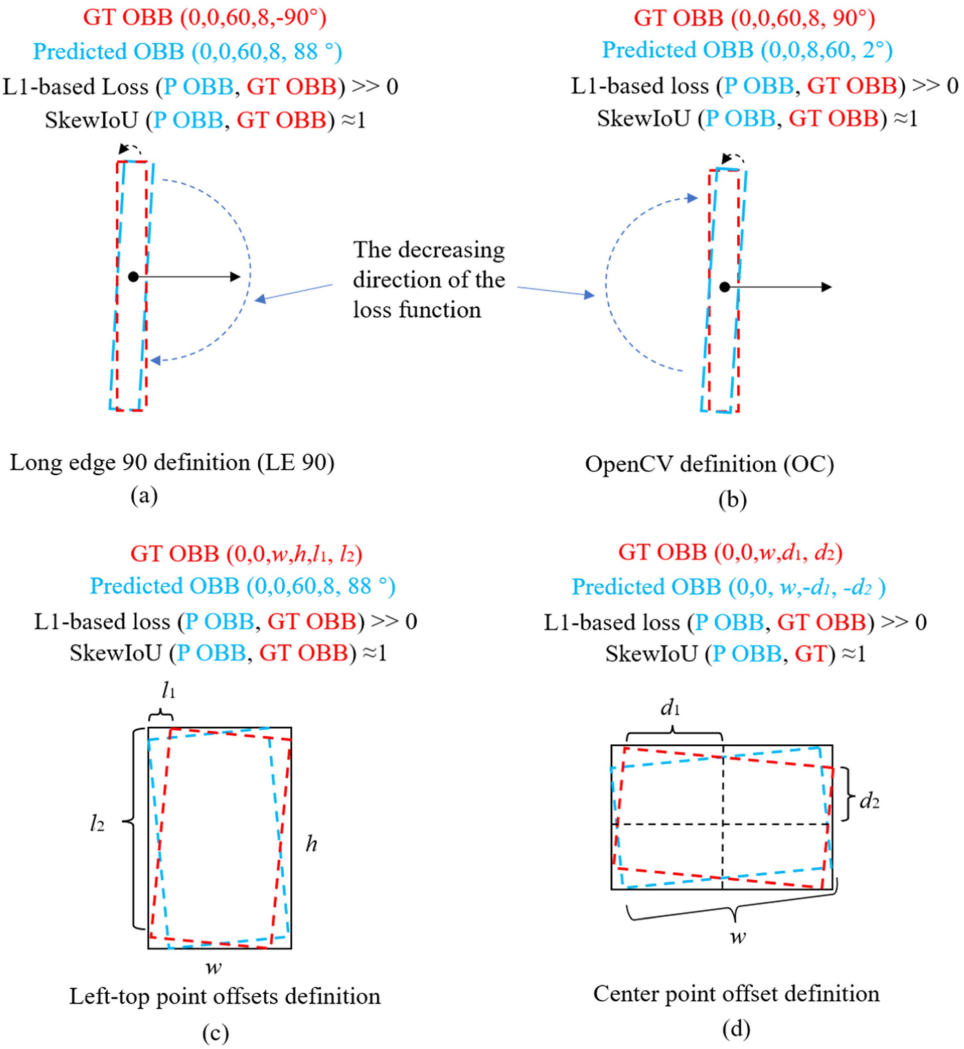


Fig. 1. Boundary discontinuity under two bounding box definitions. (a) Long edge 90 definition. (b) OpenCV definition. (c) Left-top point offsets definition [13]. (d) Center point offset definition [14].

OBBs. In [13] and [14], two point-based offsets are used to represent OBB. However, these two methods also have the problem of loss and metric inconsistency, as shown in Fig. 1(c) and (d).

Currently, the cognition of the boundary discontinuity problem is limited to the inconsistency of loss and metric. As a result, many new loss functions are proposed to replace the L1-based loss function. Researchers have proposed different loss functions to fit SkewIoU, so that the loss and the metric have better consistency. These methods are called joint optimization [17], [18], [19], [23], [24], [25], [26]. In GWD loss proposed in [23], a two-dimensional Gaussian distribution is first used to model the OBB of the rotated object. Then, the distance between the two-dimensional Gaussian distributions is calculated using GWD as an approximation of SkewIoU. Similarly, the Kullback–Leibler divergence (KLD) loss [26] models the OBB using the same method and then computes the KLD between the two-dimensional Gaussian distributions to simulate the variations of SkewIoU. In general, joint optimization is to treat OBB parameters as a whole and use the loss function to fit SkewIoU variation. Due to the high consistency between joint optimization loss and metric, models trained by joint loss functions generally perform better than independent optimization.

However, even the state-of-the-art joint optimization loss function cannot solve the problem of boundary discontinuity. To illustrate this, with the encoding method as LE90, we use the KLD joint loss function [26] to train the fully convolutional one-stage (FCOS) network [27] to get the detection model. With different rotated angles of the input SAR ship image, the model predicted angles are shown in Fig. 2(a) (a polar coordinate system, the angular coordinate represents the rotation angle of ship, the radial coordinate indicates the angle value), where the blue line represents the desired predicted angle and the red dots represent the actual predicted angle. It can be seen that the actual predicted angle is very close to the desired angle in most cases, but there is a large error near the boundary angles ($\pi/2$ and $3\pi/2$). The reason for this problem is that the model tries to fit the angle relationship between the box and the target during training, but this relationship is essentially a piecewise function

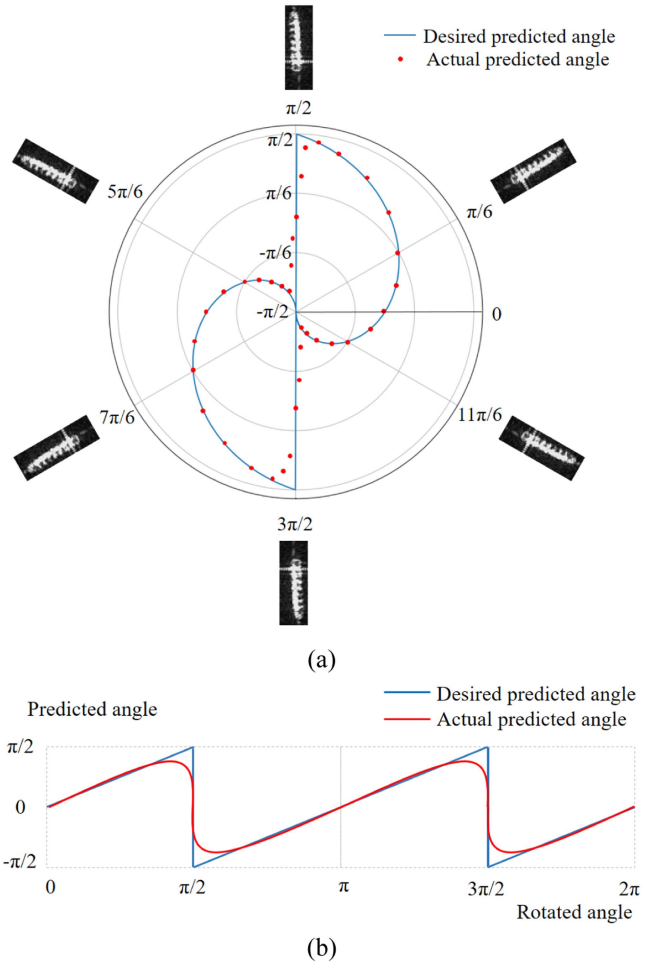


Fig. 2. Desired predicted angle and the actual predicted angle. (a) Polar coordinate system. (b) Cartesian coordinate system.

with breakpoints, as shown in Fig. 2(b). Many research works have shown that neural networks have strong fitting capabilities for continuous functions, while their fitting capability for discontinuous functions is relatively poor [28], [29]. This can lead to abrupt changes and instability at the points of discontinuity [30]. As a result, the model's actual output is closer to the red curve in Fig. 2(b) rather than the blue curve with discontinuities, which in turn causes the issue of boundary discontinuity.

Different from LE90 definition, OC definition has prediction breakpoints of both angle and edge in the boundary angle region. In the new encoding method proposed in [13] and [14], there is also the problems of encoding discontinuity.

Because the joint optimization has better metric-loss consistency and can improve the model detection accuracy, researchers mistakenly believe that the joint optimization can solve the boundary discontinuity problem [22]. Therefore, to solve the boundary discontinuity problem, the most important step is to find a continuous encoding method to make the model fit a smooth function rather than that with some breakpoints. In addition, the joint optimization loss functions need to be defined based on this continuous encoding.

III. Proposed Method

In this section, we first present the conditions that must be met to address the issue of boundary discontinuity. To fulfill these conditions, we propose a continuous encoding method called CDM and a loss function based on joint optimization. We also provide a design paradigm for the detection model that utilizes CDM and the joint optimization function.

A. Conditions of Boundary Continuity

In the field of oriented SAR ship detection, as described in Fig. 1, the commonly used OBB representation methods often face the challenge of boundary discontinuity. In addition, the study of the loss function is often a struggle against its sudden change, which does not solve the problem of boundary discontinuity fundamentally.

In order to define continuous encoding in theory, we give the following definition. We define $E(x)$ as the encoding function from the SAR ship target x to the OBB parameter space $\mathbb{R}^n$, where n indicates the number of parameters. Define $T(x;\theta)$ as the transformation that rotates x counterclockwise by an angle of θ and S as the set of the state spaces of x. In order to ensure the encoding continuity (EC), it is necessary to satisfy that slight rotation should affect OBB parameters to a minimum, that is

$$\forall x \in S, \lim_{\theta \to 0} ||E(T(x;\theta)) - E(x)|| = 0. \tag{1}$$

Define $D()$ as the decoding function and p as a set of OBB parameters in $\mathbb{R}^n$. In addition to satisfying EC, decoding needs to satisfy decoding completeness (DC) to ensure that every state of x can be accurately represented, that is

$$\forall x \in S, \exists p \in \mathbb{R}^n, \mathrm{SkewIoU}(x, D(p)) = 1. \tag{2}$$

Define $L_{\mathrm{E}}()$ and $L_{\mathrm{D}}()$ to represent the loss functions calculated using the encoding parameters and decoding parameters, respectively. During the rotation of the ship target, the loss function should not exhibit abrupt changes. In other words, the loss function should possess rotation continuity, that is

$$\begin{aligned} &\forall x \in S, \lim_{\theta \to 0} ||L_E(E(T(x;\theta)), E(x))|| = 0 \\ &\text{and} \lim_{\theta \to 0} ||L_D(D(E(T(x;\theta))), D(E(x)))|| = 0. \end{aligned} \tag{3}$$

In the case of using only encoding parameters or only decoding parameters, it is sufficient to meet the corresponding condition. Here, we refer to the capabilities corresponding to (1), (2), and (3) as EC, DC and loss rotation continuity (LRC) respectively. The OBB encoding of typical existing methods in the field of oriented SAR ship detection could not have these three capabilities at the same time, as shown in Table I. In view of this, we propose an oriented SAR ship detection framework to solve the problem of boundary discontinuity. This framework includes the design of the OBB encoding and the design of the loss function.

TABLE I
OBB ENCODING CAPABILITIES OF TYPICAL METHODS

Methods	EC	DC	LRC
LTPO [14]	×	√	×
FPDDet [15]	×	√	×
Reference [19]	×	√	√
NPA2Net [31]	×	×	×
SPG-OSD [32]	×	√	×
Reference [33]	×	√	×

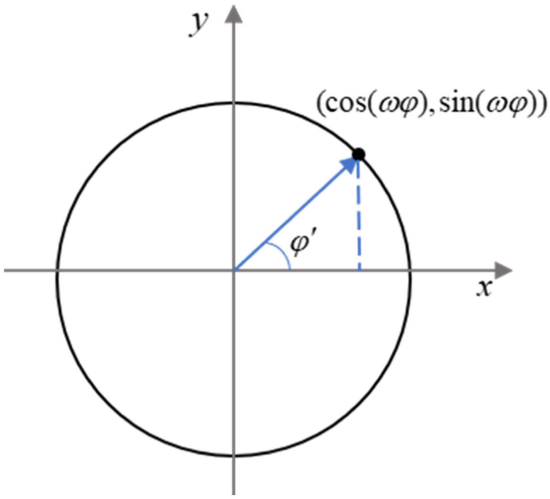


Fig. 3. Coordinate representation of a point on the unit circle in the Cartesian coordinate system.

B. Continuous Encoding Method

How to realize the continuous encoding of OBB is the key to solve the boundary discontinuity problem. Here, a continuous encoding method based on coordinate decomposition is proposed, called CDM.

Taking the LE90 definition as an example, the parameters (cx, cy, w, h) maintain periodic invariance during the rotation of the object. The source of boundary discontinuity stems from the periodicity and discontinuity of the OBB angle. Therefore, finding a way to encode the angle of the OBB that can fully represent it and ensures that the encoding function is continuous is key to resolving the issue of boundary discontinuity.

For ease of expression, we change the angle range of OBB in LE90 definition to $[0, \pi)$. Define θ_{obb} as the rotated angle of OBB and θ_t as the rotated angle of ship targets. The ranges of θ_{obb} and θ_t are $[0, \pi)$ and $[0, 2\pi)$, respectively. The transformation relationship between θ_{obb} and θ_t can be expressed as

$$\theta_{\text{obb}} = \begin{cases} \theta_t, & 0 \le \theta_t < \pi \\ \theta_t - \pi, & \pi \le \theta_t < 2\pi \end{cases}. \tag{4}$$

It follows that θ_{obb} has a period of π and is discontinuous at $\theta_t = 0\&\pi$, corresponding to $\pi/2\&3\pi/2$ in Fig. 2(b). To obtain continuous encoding, we need to find a function that has period π and is continuous.

As shown in Fig. 3, the coordinates of a point on the unit circle can be represented as $(\cos\omega\varphi, \sin\omega\varphi)$, while the angle φ' can be expressed as

$$\begin{aligned} \varphi' &= \frac{1}{\omega}\arctan 2(\sin\omega\varphi, \cos\omega\varphi) \\ &= \begin{cases} \varphi & 0 \le \varphi < \frac{2\pi}{\omega} \\ \varphi \bmod \frac{2\pi}{\omega} & \varphi \ge \frac{2\pi}{\omega} \end{cases} \end{aligned} \tag{5}$$

where mod represents the modulus operation. The arctan2 function is a quadrant function that can calculate the radian information of the point on the Cartesian coordinate axis, and its range is $[0, 2\pi)$. Its mathematical expression is as

$$\arctan 2(y, x) = \begin{cases} \arctan\frac{y}{x} + \pi & x < 0 \\ \arctan\frac{y}{x} & x > 0, y \ge 0 \\ \arctan\frac{y}{x} + 2\pi & x > 0, y < 0 \\ \frac{\pi}{2} & x = 0, y > 0 \\ \frac{3\pi}{2} & x = 0, y < 0 \\ \text{undefined} & x = 0, y = 0 \end{cases}. \tag{6}$$

When $\omega = 2$, (5) becomes

$$\begin{aligned} \varphi' &= \frac{1}{2}\arctan 2(\sin 2\varphi, \cos 2\varphi) \\ &= \begin{cases} \varphi, & 0 \le \varphi < \pi \\ \varphi \bmod \pi, & \varphi \ge \pi \end{cases}. \end{aligned} \tag{7}$$

When the range of φ is $[0, 2\pi)$, (7) can be written as

$$\varphi' = \begin{cases} \varphi, & 0 \le \varphi < \pi \\ \varphi - \pi, & \pi \le \varphi < 2\pi \end{cases}. \tag{8}$$

In this case, (8) and (4) are equivalent, so we can take the coordinates of the points on the unit circle to encode the angle θ_{obb}. The encoding function can be expressed as

$$E_\theta(\theta_t) = (\alpha_1, \alpha_2) = (\cos 2\theta_t, \sin 2\theta_t) \tag{9}$$

where α_1 and α_2 represent encoding parameters. The decoding function can be expressed as

$$\theta_{\text{obb}} = D_\theta(E_\theta(\theta_t)) = \frac{1}{2}\arctan 2(\alpha_2, \alpha_1). \tag{10}$$

In (9) and (10), $E_\theta()$ and $D_\theta()$ represent functions that encode and decode the angle parameter of x, respectively. Since we do not encode and decode the other four parameters (cx, cy, w, h), we have written the encoding function $E()$ and decoding function $D()$ as $E_\theta()$ and $D_\theta()$ for the angle parameter only.

We draw the changes of the values of θ_{obb}, α_1 and α_2 in the polar coordinate system to more intuitively explain the relationship between θ_{obb}, α_1, α_2, and θ_t, as shown in Fig. 4, where the azimuth axis represents the rotated angle θ_t of the target and the radial axis represents the magnitude of the corresponding value. It can be seen that as the azimuth angle rotates, α_1 and α_2 remain continuous and there is no discontinuity point. Therefore, *the encoding function can satisfy EC*. In addition, it can also be seen from Fig. 4 that for any rotation angle θ_{obb}, there is always a pair (α_1, α_2) corresponding to it, which guarantees the *completeness of the decoding function*.

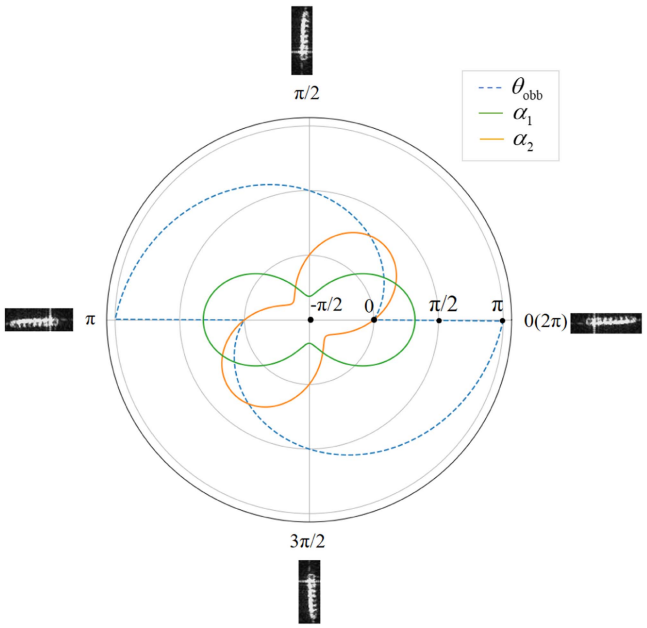


Fig. 4. Schematic diagram of the variation of $\theta_{\rm obb}$, α_1 and α_2 with the change of the target rotation angle θ_t in the polar coordinate system.

C. Joint Optimization Loss

On the basis of the continuous coding method CDM, it is necessary to design a suitable optimization loss function. In this section, we propose a joint optimization loss based on CDM.

In the field of oriented target detection, there are two main loss optimization paradigms, independent optimization and joint optimization.

Independent optimization: This type of method optimizes each encoding parameter individually. For example, using the Smooth L1 loss to calculate the difference between each predicted parameter and the GT parameter, or dividing the predicted results into a HBB (cx, cy, w, h) and a rotation angle θ, where box regression uses HBB loss (e.g., GIoU loss [34] or DIoU loss [35]) and angle regression uses L1 loss. These methods often encounter issues such as inconsistency between loss and metrics, as well as sudden changes in loss value, as shown in Fig. 1.

Joint optimization: This type of method decodes the predicted parameters into predicted OBBs and then calculates the SkewIoU-based loss (e.g., GWD loss [23] and KLD loss [26]) between the predicted OBBs and the GT OBBs. Such methods typically demonstrate better performance because they are more consistent with the final evaluation metrics, which has already reached a consensus. In addition, because the joint optimization method does not calculate the angle loss directly, it can avoid the problem of sudden change of the loss function caused by the PoA.

Actually, no matter joint optimization or independent optimization functions, if they are based on a continuous encoding, they can become continuous loss functions. For example, the smooth L1 loss function becomes discontinuous due to the boundary discontinuity when it is based on the LE90 method. If based on the CDM method, this loss function becomes continuous because the encoding is continuous. However, it cannot be denied that joint optimization has an advantage in detection accuracy compared to independent optimization. Hence, we also adopt the form of loss function based on joint optimization, where the joint optimization loss function $L_{\rm obb}$ is expressed as

$$L_{\rm obb} = L_{\rm SkewIoU}(B_p(cx_p, cy_p, w_p, h_p, \theta_p), B_{\rm GT}(cx_t, cy_t, w_t, h_t, \theta_t)) \tag{11}$$

where B_p represents the predicted OBB, whose parameters are cx_p, cy_p, w_p, h_p and θ_p, $B_{\rm GT}$ denotes the GT OBB, whose parameters are cx_t, cy_t, w_t, h_t and θ_t, (cx, cy) indicates the center position, (w, h) represents the width and height, and θ represents the rotated angle, $L_{\rm SkewIoU}$ refers to the SkewIoU-based loss function, such as KLD loss, GWD loss, and KFIoU loss [17], and θ_p is obtained by decoding the encoding parameters, expressed as

$$\theta_p = D_\theta(\alpha_1^p, \alpha_2^p) \tag{12}$$

where α_1^p and α_2^p represent the predicted encoding parameters.

In order to better constrain the encoding parameters, we also add an angle encoding loss $L_{\rm ang}$ to achieve faster convergence, expressed as

$$L_{\rm ang} = L_1((\alpha_1^p, \alpha_2^p), (\alpha_1^t, \alpha_2^t)) \tag{13}$$

where α_1^t and α_2^t represent the target encoding parameters obtained by the encoding function E_θ () for the rotated angle θ_t of GT OBB and L_1 represents the L1 loss function.

The total loss $L_{\rm total}$ can be expressed as

$$L_{\rm total} = w_{\rm cls}L_{\rm cls} + w_{\rm obb}L_{\rm obb} + w_{\rm ang}L_{\rm ang} \tag{14}$$

where $L_{\rm cls}$ represents classification loss, which is the focal loss in this paper, $w_{\rm cls}$, $w_{\rm obb}$, and $w_{\rm ang}$ are the weights for $L_{\rm cls}$, $L_{\rm obb}$, and $L_{\rm ang}$, respectively. It is important to note that $L_{\rm tatal}$ is not fixed and needs to be adjusted according to the detection network. For example, the FCOS [27] network requires an additional center loss.

In (14), the joint loss function $L_{\rm obb}$, such as KLD loss, GWD loss, and KFIoU loss, meets rotation continuity; due to the continuity of the CDM encoding, $L_{\rm ang}$ also meets rotation continuity; $L_{\rm cls}$ is a classification loss that is independent of the target's rotation. Therefore, $L_{\rm total}$ satisfies the condition of *LRC required by (3).*

D. Implementation Paradigm

Our proposed method can be easily embedded into existing detection networks, and its basic network paradigm is shown in Fig. 5.

In this basic paradigm, the backbone and neck are consistent with typical detection model frameworks. The backbone can choose ResNet [36], ReResNet [37], ARC [38], etc., while the neck can select FPN [39], LR-FPN [40], and BFPN [41], among others. The only difference is the addition of an angle encoding branch in the detection head, whose output is a two-dimensional angle encoding parameters β_1 and β_2.

According to (9), the value range of the angle encoding parameters is $[-11]$. In order to make the training more stable,

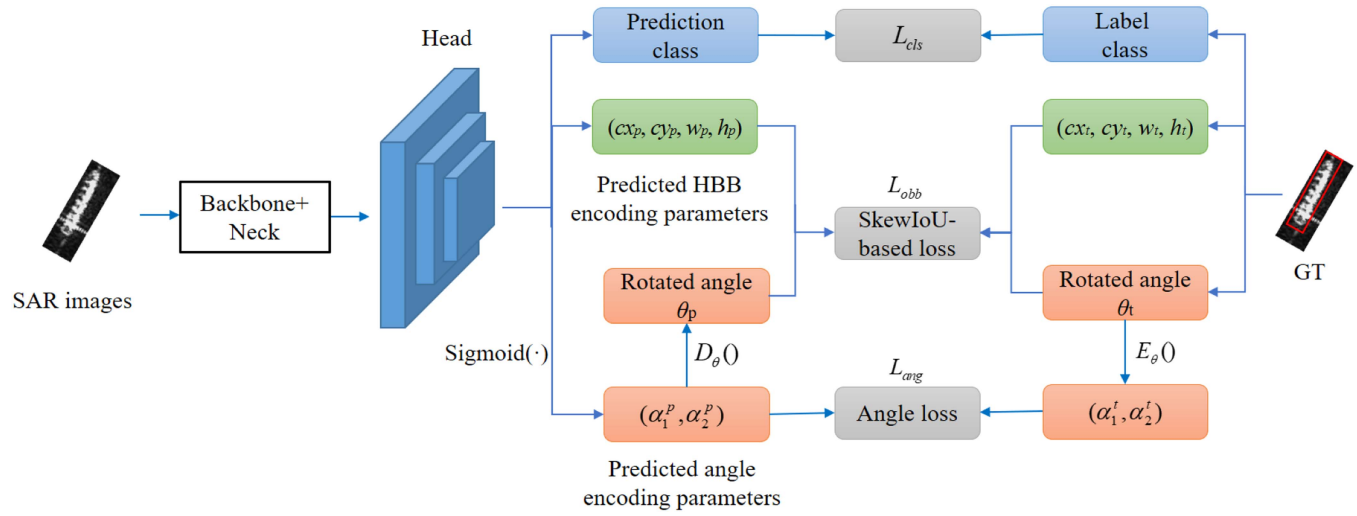


Fig. 5. Illustration of the proposed method embedded in the detection network.

the output parameters are transformed as follows to obtain the final angle encoding parameters $\alpha_1^p$and α_2^p

$$(\alpha_1^p, \alpha_2^p) = 2 \times \text{Sigmoid}(\beta_1, \beta_2) - 1. \tag{15}$$

After obtaining the predicted angle parameters, the rotated angle θ_p is decoded through the decoding function D_θ () to form a set of OBB parameters along with the predicted HBB parameters (cx_p, cy_p, w_p, h_p). Subsequently, the SkewIoU-based loss is used to calculate the loss value between predicted OBB and GT OBB. In terms of angle encoding constraints, the target encoding parameters α_1^t and $\alpha_2^t$are obtained by encoding the target angle θ_t through the encoding function E_θ (), and the L1 loss function is used to calculate the difference between the predicted parameters and the target parameters.

In this article, we refer to the network design paradigm based on the loss function (14) and the network structure based on Fig. 5 as the joint optimization paradigm.

IV. Experiment

A. Dataset and Experiment Setting

In order to verify the effectiveness of the encoding and optimization methods proposed in this article to solve the boundary discontinuity problem, experiments are carried out on two public SAR ship detection datasets.

RSDD-SAR Dataset [42]: It is a SAR ship detection dataset consists of 84 scenes of Gaofen-3 data, 41 scenes of TerraSAR-X data, and 2 untrimmed large images, i.e., 127 scenes of data in total. These data encompass various imaging modes, polarizations, and resolutions, specifically including 7000 slices and 10 263 ship instances. Among these, 5000 images contain 7271 ship objects for training, while 2000 images include 2992 ship objects for testing. The image resolution in the RSDD-SAR dataset ranges from 2 to 20 m, with a consistent image size of 512 × 512 pixels.

RSSDD dataset [43]: The RSSDD dataset consists of 1160 SAR images, featuring various resolutions, polarization modes, and sea conditions. It encompasses a range of scenarios, including near-sea scenarios with considerable land clutter and those with less clutter. Regarding the division of the training set, the default principle is adopted, comprising a training set of 928 images and a testing set of 232 images. The size of the images in this dataset ranges from 271 × 241 to 526 × 646 in various dimensions. To standardize the size, they are resized to a uniform dimension of 512 × 512 pixels.

We used the MMRotate platform [44] to complete our comparative experiment, with the learning rate as 0.0001, epoch as 36, batchsize as 4, the optimizer as Adam. All of our experiments were done on the PyTorch platform, the hardware platform is the NVIDIA GeForce RTX A6000 GPU. We have integrated our methods into the MMRotate platform to enable fast calls. More technical details can be found in the code we publish later.

B. Evaluation Metrics

To objectively evaluate the effectiveness of the method proposed in this article, we used three evaluation metrics from the MS COCO [45] dataset: AP, AP_{50}, and AP_{75}. AP is defined as follows:

$$\text{AP} = \int_0^1 P(R)dR \tag{16}$$

where $P(R)$ stands for precision recall curve (PRC), and AP is defined as the area under the PRC. The higher the AP value, the better the detection performance of the method. In the MS COCO dataset, AP is calculated according to the average accuracy of IoU values from 0.5 to 0.95, with a step size of 0.05, and the same calculation method is used in this article. P represents precision, which is the ratio of correctly detected ship samples to all predicted positive ship samples, and R stands for recall, which is the ratio of correctly detected ship samples to all labeled positive ship samples, expressed as

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}, R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

TABLE II
COMPARISON OF DETECTION ACCURACIES OF DIFFERENT ENCODING METHODS ON RSDD-SAR DATASET

Methods	Angle encoding mode	Angle encoding length	AP	AP_{50}	AP_{75}
R-RetinaNet	LE90 (Baseline)	1	44.9	85.9	43.4
R-FCOS			49.6	88.5	52.1
R-RetinaNet	CSL	180	36.8	83.7	23.9
R-FCOS			43.4	85.6	35.3
R-RetinaNet	CDM	2	47.6	88.1	47.8
R-FCOS			**51.1**	**90.9**	**53.5**

TABLE III
COMPARISON OF DETECTION ACCURACIES OF DIFFERENT ENCODING METHODS ON RSSDD DATASET

Methods	Angle encoding mode	Angle encoding length	AP	AP_{50}	AP_{75}
R-RetinaNet	LE90 (Baseline)	1	48.5	87.1	49.9
R-FCOS			51.8	88.0	59.9
R-RetinaNet	CSL	180	40.7	85.3	24.3
R-FCOS			44.8	87.5	31.9
R-RetinaNet	CDM	2	50.8	90.2	54.4
R-FCOS			**53.1**	**90.3**	**61.6**

where FN refers to false negatives, indicating samples that were incorrectly identified as negative, FP stands for false positives, which represent samples that were incorrectly identified as positive, and TP denotes true positives, referring to samples that were correctly identified as positive.

AP_{50} and AP_{75} represent the APs at SkewIoU thresholds of 0.5 and 0.75, respectively. Compared with AP_{50}, AP_{75} can better reflect the accuracy of high-precision detection of the model.

C. Comparison of Different Encoding Modes

To compare the impact of different angle encoding methods on detection performance, this article conducts comparative experiments between the CDM and the commonly used LE90 and CSL encoding methods. LE_{90} is the method illustrated in Fig. 1(a). CSL is the method proposed in [46], which primarily aims to transform the angle regression problem into an angle classification problem, thereby avoiding the issue of discontinuity at angle boundaries. In this article, the angle encoding length for CSL is set to 180, meaning that the angles of the regression boxes are divided into 180 categories, with each category representing one degree. In this section, the model is trained using an independent optimization method.

Considering that R-RetinaNet [47] and R-FCOS [27] are baseline networks for many SAR ship detection methods, this article also performed ablation experiments on these two networks. RetinaNet is the baseline network for anchor-based detection, while FCOS is the baseline network for anchor-free detection. The difference between R-RetinaNet and RetinaNet is that R-RetinaNet has an additional angle branch in the detection head to predict the orientation of OBBs, and the same applies to R-FCOS. ResNet-50 is used as the backbone network for all networks, and MMRotate's official FPN is used for Neck. These setting is followed in subsequent experiments. For RSDD-SAR and RSSDD datasets, the detection accuracies of different encoding methods are shown in Tables II and III.

It can be observed that, compared to the LE90 method, the CSL method shows a performance decline. This is primarily due to the need for a large number of training samples to accurately classify angles when they are encoded as categories. In addition, an excessively long encoding length increases training difficulty and complexity. Finally, discretizing angles into categories introduces additional discretization errors.

Our proposed CDM encoding method has only increased the encoding length by one bit, but the detection accuracy of both detection networks has been greatly improved. This is because the LE90 angle encoding method will encounter the problem of boundary discontinuity, leading to angle prediction errors and optimization difficulties at the angle boundaries. The CDM encoding is a continuous encoding method, which solves the problem of unstable angle prediction and difficult optimization at the angle boundaries.

We have also provided some detection results of SAR ship targets, as shown in Fig. 6. It can be observed that the detection network using the LE90 encoding method exhibits significant angle prediction errors when the GT OBB of the ship target is around $\pm 90°$, which is consistent with our analysis. Due to the discontinuity of the boundaries, the network exhibits unstable predictions at the angle boundaries, leading to incorrect angle predictions.

After employing the CDM, the issues of angle prediction errors are significantly mitigated, indicating that the CDM continuous encoding method can effectively address the problem of boundary discontinuity in the orientation ship detection. Since the CDM method is continuous at angle boundary, there will be no abrupt changes in angle values at the boundaries, which helps avoid instability in angle predictions.

D. Comparison of Different Optimization Methods

The results shown in Tables II and III are the outcomes of independent optimization. Due to the higher metric consistency of the joint optimization method compared to independent optimization, it has become the mainstream approach for SAR ship detection. In this section, we analyze the impact of combining the CDM encoding method with joint optimization on detection networks. The experiments in this section are also based on the RSDD-SAR and RSSDD datasets.

To determine the value of the angle loss weight w_{ang} in (14), we analyze the detection accuracies of the detection network on the RSDD-SAR dataset under different w_{ang} settings, as shown in Table IV. Here, L_{obb} selects KLD loss, L_{cls} selects Focal loss, and we set both w_{cls} and w_{obb} to 1 in accordance with the settings in [26]. The results in Table IV indicate that when $w_{\text{ang}} = 0.1$ and $w_{\text{ang}} = 0.15$, R-RetinaNet and R-FCOS achieved the best detection accuracy, respectively. Therefore, in subsequent experiments, weights of 0.1 and 0.15 are used for the

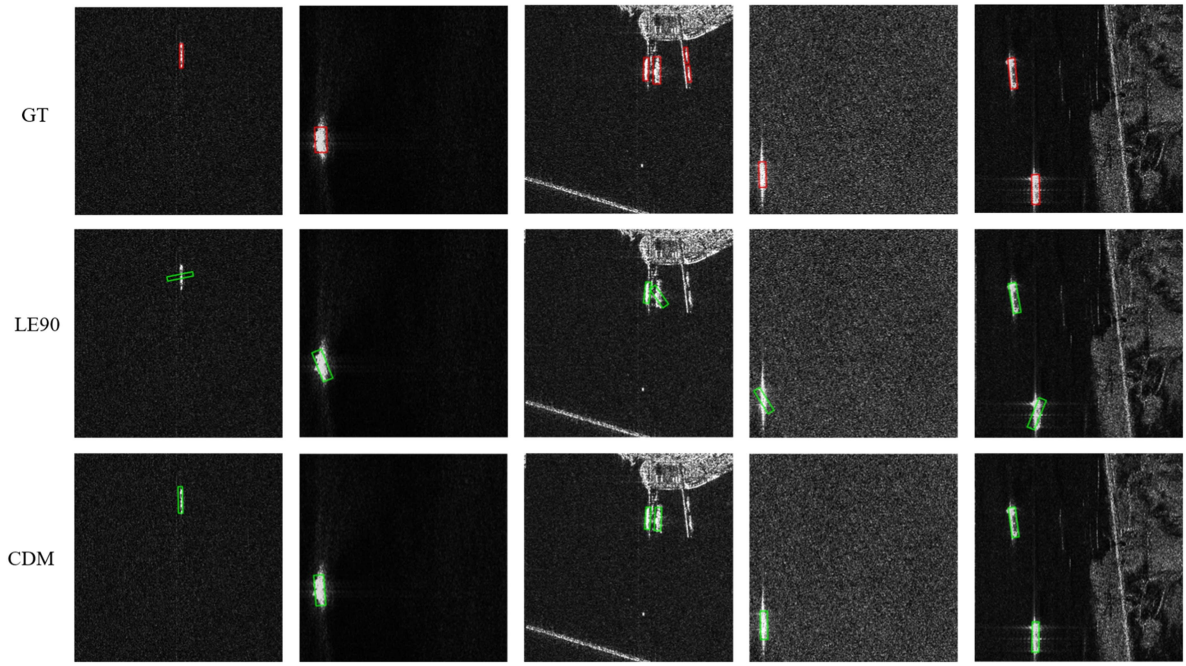


Fig. 6. Detection results of SAR images for ship targets using different encoding methods (the red box represents the GT OBB and the blue box represents the predicted OBB. The first line, the second line, and the third line correspond to the GT label, the prediction results of the LE90 encoding method, and the prediction results of the CDM, respectively).

TABLE IV
DETECTION ACCURACIES OF DIFFERENT MODELS UNDER DIFFERENT ANGLE LOSS WEIGHTS

w_{ang}	R-RetinaNet		R-FCOS	
	AP_{50}	AP_{75}	AP_{50}	AP_{75}
0.00	85.7	42.0	89.1	48.7
0.05	89.3	48.4	90.8	54.6
0.10	**89.5**	**48.7**	91.1	54.8
0.15	89.0	48.3	**91.7**	**55.0**
0.20	88.9	48.2	91.0	54.9
0.40	88.4	48.0	90.7	54.6
0.60	88.3	48.0	90.4	54.2

angle loss in these two networks. In addition, it is important to note that when $w_{\text{ang}} = 0$, both networks experienced a significant performance decline, indicating that angle loss is essential in the loss function.

Tables V and VI show the ablation study on RSDD-SAR and RSSDD datasets with different optimization methods, where we denote independent optimization as IO and joint optimization as JO. The independent optimization loss for R-RetinaNet is the Smooth L1 loss between the predicted OBB parameters (cx_{p}, cy_{p}, w_{p}, h_{p}, θ_{p}) and the GT OBB parameters (cx_{t}, cy_{t}, w_{t}, h_{t}, θ_{t}). The independent optimization loss for R-FCOS is the sum of the HBB loss and the angle loss, where the HBB loss is the IoU loss and the angle loss is the L1 loss. In the joint optimization of the two detection networks, we selected GWD Loss and KLD Loss as L_{obb}, and the implementation method is illustrated in Fig. 5.

Both GWD Loss and KLD Loss aim to replace the SkewIoU loss by approximating the trend of changes in the SkewIoU values. As shown in Tables V and VI, the joint optimization method using GWD Loss and KLD Loss significantly improves the detection accuracy of these two detection networks across both datasets compared to independent optimization. However, the joint optimization loss still does not fundamentally resolve the issue of discontinuity at angular boundaries (as shown in Fig. 2). When GWD Loss and KLD Loss are combined with the CDM method, the detection accuracy is further enhanced. This indicates that the encoding method of CDM can also improve the performance of the detector in joint optimization.

E. Comparison With the State-of-the-Art Methods

Table VII shows the results of comparison with the existing target detection methods on two datasets, which are mainly divided into three categories.

1) Two-Stage: Two-stage detection methods split the detection process into two phases: first generating candidate boxes, and then classifying and detecting the objects based on the features of these candidate boxes. This category mainly includes Faster R-CNN [48], ReDet [37], and Oriented R-CNN [49].

2) Single-Stage: This category of detection networks directly outputs detection results, offering the advantage of fast detection rates. Due to the high-speed requirements for SAR ship detection, single-stage detection networks are commonly used to detect ship targets. This category mainly includes S2ANet [50], O-Reppoints [51], and R3Det [24].

3) Encoding-Based: These methods aim to improve detection performance by modifying the encoding methods of the oriented boxes. This category primarily includes Gliding Vertex [52], PSC [53], EPE [8], LTPO [14], FPDDet [15] and YOLO11

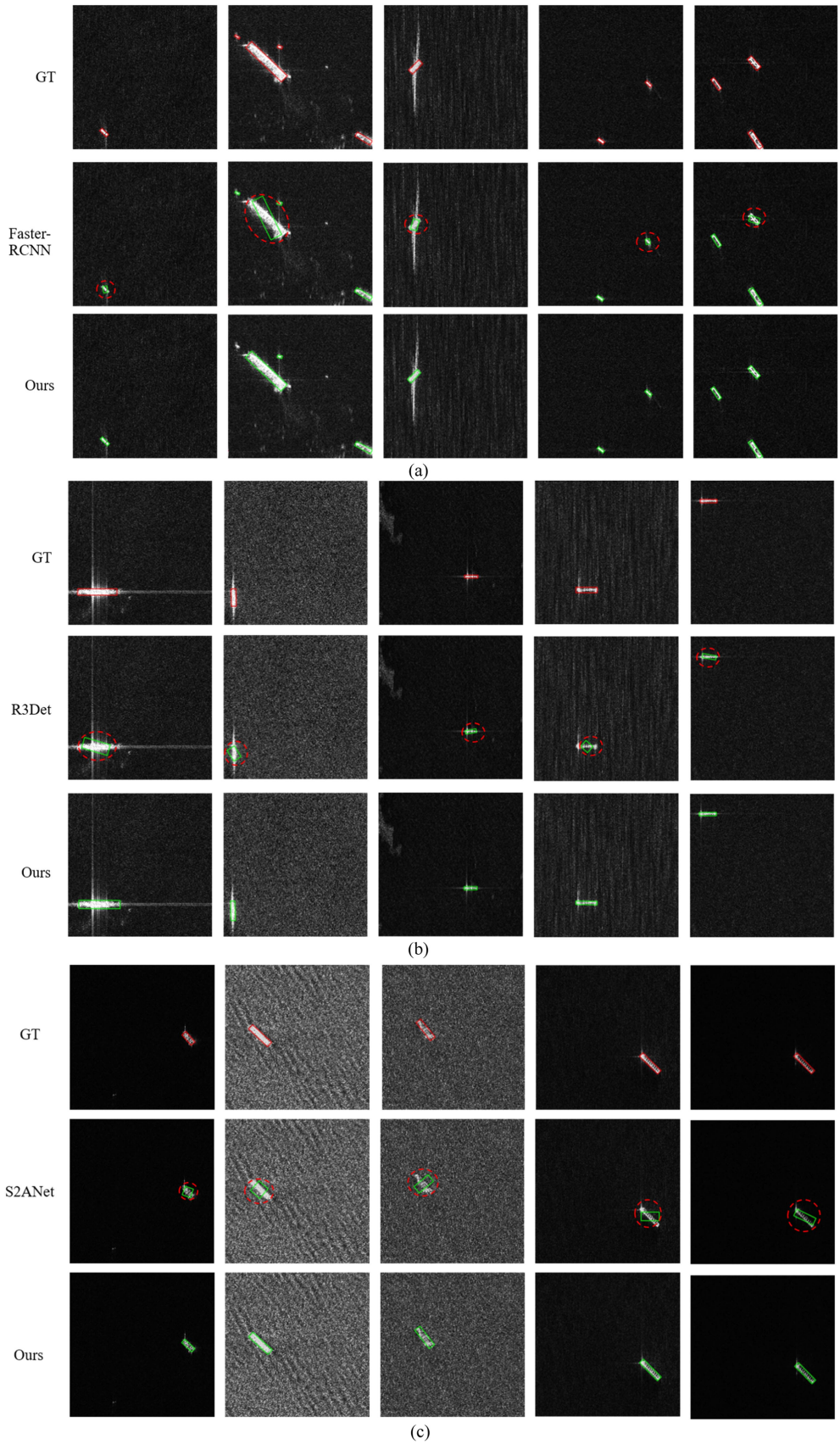


Fig. 7. Detection result comparisons between our network and (a) Faster-RCNN, (b) R3Det, and (c) S2ANet. The dotted circles indicate the incorrect angle predictions.

TABLE V
ABLATION STUDY ON RSDD-SAR DATASET WITH DIFFERENT OPTIMIZATION METHODS

Methods	Optimization mode	Method	AP	AP_{50}	AP_{75}
R-RetinaNet	IO	Baseline	44.9	85.9	43.4
		CDM	47.6(+2.7)	88.1(+2.2)	47.8(+4.4)
	JO	GWD	47.7(+2.8)	88.2(+2.3)	47.7(+4.3)
		GWD + CDM	48.1(+3.2)	89.2(+3.3)	48.6(+5.2)
		KLD	47.9(+3.0)	88.7(+2.8)	48.1(+4.7)
		KLD + CDM	48.2(+3.3)	89.5(+3.6)	48.7(+5.3)
R-FCOS	IO	Baseline	49.6	88.5	52.1
		CDM	51.1(+1.5)	90.9(+2.4)	53.5(+1.4)
	JO	GWD	50.9(+1.3)	90.9(+2.4)	52.8(+0.7)
		GWD + CDM	51.4(+1.8)	91.1(+2.6)	54.6(+2.5)
		KLD	51.0(+1.4)	91.0(+2.5)	52.8(+0.7)
		KLD + CDM	51.5(+1.9)	91.7(+3.2)	55.0(+2.9)

TABLE VI
ABLATION STUDY ON RSSDD DATASETS WITH DIFFERENT OPTIMIZATION METHODS

Networks	Optimization mode	Method	AP	AP_{50}	AP_{75}
R-RetinaNet	IO	Baseline	48.5	87.1	49.9
		CDM	50.8(+2.3)	90.2(+3.1)	54.4(+4.5)
	JO	GWD	51.2(+2.7)	90.3(+3.2)	56.4(+6.5)
		GWD + CDM	51.9(+3.4)	91.2(+4.1)	58.6(+8.7)
		KLD	51.7(+3.2)	90.7(+3.6)	57.1(+7.2)
		KLD + CDM	52.2(+3.7)	91.9(+4.8)	58.7(+8.8)
R-FCOS	IO	Baseline	51.8	88.0	59.9
		CDM	53.1(+1.3)	90.3(+2.3)	61.6(+1.7)
	JO	GWD	52.3(+0.5)	90.0(+2.0)	60.5(+0.6)
		GWD + CDM	53.2(+1.4)	91.9(+3.9)	62.6(+2.7)
		KLD	54.5(+2.7)	94.0(+6.0)	62.4(+2.5)
		KLD + CDM	55.3(+3.5)	95.3(+7.3)	63.8(+3.9)

TABLE VII
DETECTION RESULTS OF DIFFERENT METHODS ON RSDD-SAR AND RSSDD

Type	Methods	RSDD-SAR			RSSDD		
		AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
Two-stage	Faster R-CNN [48]	49.2	89.2	50.7	51.3	87.7	54.9
	ReDet [37]	49.8	90.5	51.1	50.1	89.5	52.7
	Oriented R-CNN [49]	49.7	89.4	51.9	52.9	91.4	56.6
Single-stage	S2ANet [50]	50.2	90.7	53.6	53.1	93.7	59.4
	O-Reppoints [51]	49.7	89.4	51.0	52.7	92.2	58.3
	R3Det [24]	48.4	89.3	48.2	51.4	89.1	57.6
Encoding-based	Gliding Vertex [52]	49.1	89.7	48.3	54.4	93.7	61.7
	PSC [53]	50.7	91.4	53.4	54.1	94.6	62.4
	EPE* [8]	/	89.4	/	/	/	/
	LTPO*[14]	/	/	/	/	90.3	42.3
	FPDDet*[15]	/	91.1	/	/	97.8	/
	R-FCOS+KLD+CDM	**51.5**	**91.7**	**55.0**	**55.3**	95.3	**63.8**

*The results are taken from the corresponding papers.

[54]. Due to EPE, LTPO and FPDDet lack publicly available code, their results in Table VII are taken from the original papers.

It is important to note that different detection methods utilize various backbone networks, necks, detection heads, and training strategies, all of which can impact the final detection performance (e.g., the backbone of ReDet is the rotation-invariant ReResNet, while FPDDet's neck employs a feature enhancement module). Establishing a completely fair comparison among different methods becomes challenging. Despite these challenges, our "FCOS+KLD+CDM" approach achieved the best detection results among these methods on RSDD-SAR dataset, with AP_{75} of 91.7% and AP_{50} of 55.0%. On the RSSDD dataset, our model achieves the highest AP_{75} accuracy of 63.8%.

Fig. 7 shows the boundary discontinuities in some models and the comparison with our model. The boundary angles of different models are different. It can be seen that there are errors in angle prediction caused by boundary discontinuity in many models. The root cause is the existence of discontinuous OBB encoding. Our approach solves the problem of incorrect prediction of boundary angles.

V. Discussion

It is important to note that this article does not discuss the issue of boundary discontinuity in square-like objects detection. This is primarily because the majority of ship targets have a large aspect ratio, and square-like objects are rare in SAR ship detection. Therefore, the boundary discontinuity of square-like objects has little effect on oriented SAR ship detection.

In Figs. 6 and 7, for ship targets that are at the boundary angles, some may experience incorrect angle predictions when detected using the LE90 encoding method, while others may not. This is because the use of non-maximum suppression (NMS) to filter outputs can eliminate boxes affected by boundary issues. Since multiple anchor boxes (or central points) are responsible for the same object, some of them may be less affected by boundary problems. NMS can select the predicted OBB that is closest to the GT OBB. The result in Fig. 2 shows the prediction from a fixed detection location without NMS operation.

VI. Conclusion

In this article, we reveal through analysis that existing oriented SAR ship detection methods generally suffer from the issue of boundary discontinuity. To address this problem, we first outline the theoretical requirements that must be met to resolve boundary discontinuity. Furthermore, guided by these theoretical requirements, we propose a continuous encoding method called CDM. Given that joint optimization outperforms independent optimization, we introduce a joint optimization paradigm based on CDM. Our experimental results demonstrate that the CDM encoding method can effectively resolve the boundary discontinuity issue in detection and enhance the detection accuracy of the network. We applied our proposed joint optimization paradigm to R-RetinaNet and R-FCOS, conducting experimental validation on two datasets. The results indicate that this joint optimization paradigm can further improve the performance of the detection network and achieve competitive advantages in comparison with other methods without employing any additional techniques. The method in this article can be easily embedded into other detection methods, and can provide a way to solve the boundary discontinuity problem in oriented SAR ship detection. In our experiments, we also found that detection networks have a relatively high false negative rate due to strong background clutter in SAR images. Reducing the false negative rate and enhancing ship features will be the focus of our future work.

References

[1] A. H. Oveis, E. Giusti, S. Ghio, and M. Martorella, "A survey on the applications of convolutional neural networks for synthetic aperture radar: Recent advances," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 37, no. 5, pp. 18–42, May 2022.

[2] R. Gens, "Oceanographic applications of SAR remote sensing," *GI-Science Remote Sens.*, vol. 45, no. 3, pp. 275–305, 2008, doi: 10.2747/1548-1603.45.3.275.

[3] M. X. Zhang and Y. Xie, "FEC: A feature fusion framework for SAR target recognition based on electromagnetic scattering features and deep CNN features," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2174–2187, Mar. 2021, doi: 10.1109/tgrs.2020.3003264.

[4] A. C. Mondini, F. Guzzetti, K. -T. Chang, O. Monserrat, T. R. Martha, and A. Manconi, "Landslide failures detection and mapping using synthetic aperture radar: Past, present and future," *Earth- Sci. Rev.*, vol. 216, 2021, Art. no. 103574, doi: 10.1016/j.earscirev.2021.103574.

[5] D. An, W. Wang, and Z. Zhou, "Refocusing of ground moving target in circular synthetic aperture radar," *IEEE Sensors J.*, vol. 19, no. 19, pp. 8668–8674, Oct. 2019, doi: 10.1109/JSEN.2019.2922649.

[6] X. Ma, S. Hou, Y. Wang, J. Wang, and H. Wang, "Multiscale and dense ship detection in SAR images based on key-point estimation and attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5221111, doi: 10.1109/TGRS.2022.3141407.

[7] F. Ma, F. Zhang, D. Xiang, Q. Yin, and Y. Zhou, "Fast task-specific region merging for SARimage segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5222316, doi: 10.1109/TGRS.2022.3141125.

[8] J. Liu, L. Liu, and J. Xiao, "Ellipse polar encoding for oriented SAR ship detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3502–3515, 2024, doi: 10.1109/jstars.2024.3352098.

[9] F. Meng, X. Qi, and H. Fan, "LSR-det: A lightweight detector for ship detection in SAR images based on oriented bounding box," *Remote Sens.*, vol. 16, no. 17, 2024, Art. no. 3251, doi: 10.3390/rs16173251.

[10] B. Chen, C. Yu, S. Zhao, and H. Song, "An anchor-free method based on transformers and adaptive features for arbitrarily oriented ship detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2012–2028, 2024, doi: 10.1109/jstars.2023.3325573.

[11] Z. Pan, R. Yang, and Z. Zhang, "MSR2N: Multi-stage rotational region based network for arbitrary-oriented ship detection in SAR images," *Sensors*, vol. 20, no. 8, 2020, Art. no. 2340, doi: 10.3390/s20082340.

[12] Z. Sun, X. Leng, Y. Lei, B. Xiong, K. Ji, and G. Kuang, "BiFA-YOLO: A novel YOLO-based method for arbitrary-oriented ship detection in high-resolution SAR images," *Remote Sens.*, vol. 13, no. 21, 2021, Art. no. 4209.

[13] Y. Qiao, L. Miao, Z. Zhou, and Q. Ming, "A novel object detector based on high-quality rotation proposal generation and adaptive angle optimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617715, doi: 10.1109/tgrs.2023.3301610.

[14] P. Guo, T. Celik, N. Liu, and H. -C. Li, "Break through the border restriction of horizontal bounding box for arbitrary-oriented ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, 2023, Art. no. 4005505, doi: 10.1109/lgrs.2023.3270897.

[15] M. Ju, B. Niu, and J. Zhang, "FPDDet: An efficient rotated sar ship detector based on simple polar encoding and decoding," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5218915, doi: 10.1109/tgrs.2023.3324355.

[16] Z.-K. Xiao, G.-Y. Yang, X. Yang, T.-J. Mu, J. Yan, and S.-M. Hu, "Theoretically achieving continuous representation of oriented bounding boxes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 16912–16922, doi: 10.1109/CVPR52733.2024.01600.

[17] X. Yang et al., "The KFIoU loss for rotated object detection," in *Proc. 11th Int. Conf. Learn. Representations*, 2023, pp. 1–17.

[18] Z. Xu, R. Gao, K. Huang, and Q. Xu, "Triangle distance IoU Loss, attention-weighted feature pyramid network, and rotated-SAR ship dataset for arbitrary-oriented SAR ship detection," *Remote Sens.*, vol. 14, no. 18, 2022, Art. no. 4676, doi: 10.3390/rs14184676.

[19] C. Xu, H. Su, L. Gao, J. Wu, and W. Yan, "Rotated SAR ship detection based on Gaussian Wasserstein distance loss," *Mobile Netw. Appl.*, vol. 28, no. 5, pp. 1842–1851, 2023, doi: 10.1007/s11036-023-02207-6.

[20] C. Zhang, G. Gao, J. Liu, and D. Duan, "Oriented ship detection based on soft thresholding and context information in SAR images of complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5200615, doi: 10.1109/tgrs.2023.3340891.

[21] B. He, Q. Zhang, M. Tong, and C. He, "Oriented ship detector for remote sensing imagery based on pairwise branch detection head and SAR feature enhancement," *Remote Sens.*, vol. 14, no. 9, 2022, Art. no. 2177, doi: 10.3390/rs14092177.
[22] H. Xu et al., "Rethinking boundary discontinuity problem for oriented object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 17406–17415, doi: 10.1109/CVPR52733.2024.01648.
[23] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. 38th Int. Conf. Mach. Learn., Proc. Mach. Learn. Res.*, 2021, pp. 1–12.
[24] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171.
[25] X. Yang, L. Hou, Y. Zhou, W. Wang, and J. Yan, "Dense label encoding for boundary discontinuity free rotation detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 15814–15824, doi: 10.1109/CVPR46437.2021.01556.
[26] X. Yang et al., "Learning high-precision bounding box for rotated object detection via Kullback-Leibler divergence," *Adv. Neural Inf. Process. Syst.*, vol. 34, pp. 18381–18394, 2021.
[27] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 9626–9635.
[28] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Math. Control, Signals Syst.*, vol. 2, no. 4, pp. 303–314, 1989.
[29] Z. Lu et al., "The expressive power of neural networks: A view from the width," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, Paper 30.
[30] N. Rahaman et al., "On the spectral bias of neural networks," in *Proc. Int. Conf. Mach. Learn.*, May 2019, pp. 5301–5310.
[31] C. Zhang, P. Liu, H. Wang, and Y. Jin, "NPA2Net: A nested path aggregation attention network for oriented SAR ship detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9772–9789, 2024, doi: 10.1109/JSTARS.2024.3392433.
[32] Y. Zhang, D. Lu, X. Qiu, and F. Li, "Scattering-point-guided RPN for oriented ship detection in SAR images," *Remote Sens.*, vol. 15, no. 5, 2023, Art. no. 1411, doi: 10.3390/rs15051411.
[33] C. Zhang, G. Gao, J. Liu, and D. Duan, "Oriented ship detection based on soft thresholding and context information in sar images of complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5200615, doi: 10.1109/tgrs.2023.3340891.
[34] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 658–666, doi: 10.1109/CVPR.2019.00075.
[35] Z. Zheng et al., "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 12993–13000.
[36] H. K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.
[37] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 2785–2794, doi: 10.1109/CVPR46437.2021.00281.
[38] Y. Pu et al., "Adaptive rotated convolution for rotated object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 6566–6577, doi: 10.1109/ICCV51070.2023.00606.
[39] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 936–944, doi: 10.1109/CVPR.2017.106.
[40] H. Li, R. Zhang, Y. Pan, J. Ren, and F. Shen, "LR-FPN: Enhancing remote sensing object detection with location refined feature pyramid network," in *Proc. Int. Joint Conf. Neural Netw.*, 2024, pp. 1–8, doi: 10.1109/IJCNN60899.2024.10650583.
[41] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 821–830, doi: 10.1109/CVPR.2019.00091.
[42] C. Xu et al., "RSDD-SAR: Rotated ship detection dataset in SAR images," *J. Radars*, vol. 11, no. 4, pp. 581–599, 2022.
[43] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, Sep. 2021, Art. no. 3690.
[44] Y. Zhou et al., "MMRotate: A rotated object detection benchmark using PyTorch," 2022, *arXiv:2204.13317*.
[45] T. Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
[46] X. Yang and J. Yan, "On the arbitrary-oriented object detection: Classification based approaches revisited," *Int. J. Comput. Vis.*, vol. 130, pp. 1340–1365, 2022.
[47] T.-Y. Ross and G. K. H. P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2999–3007.
[48] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017, doi: 10.1109/TPAMI.2016.2577031.
[49] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 3500–3509, doi: 10.1109/ICCV48922.2021.00350.
[50] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511, doi: 10.1109/TGRS.2021.3062048.
[51] Z. Yang et al., "Reppoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 9656–9665.
[52] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021, doi: 10.1109/TPAMI.2020.2974745.
[53] Y. Yu and F. Da, "Phase-shifting coder: Predicting accurate orientation in oriented object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 13354–13363.
[54] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.



Peng Li received the B.S. degree in electrical engineering from Air Force Engineering University, Xi'an, China, in 2020, where he is currently working toward the Ph.D. degree in electronic science and technology.

His research interests include SAR target detection and recognition, SAR image understanding, and deep learning.



Cunqian Feng received the Ph.D. degree in electromagnetic fields and microwave technology from Air Force Engineering University, Xi'an, China, in 2003.

He is currently a Professor with Air Force Engineering University. His research interests include radar signal processing and electronic countermeasure.



Weike Feng was born in 1992. He received the B.E. degree in radar engineering from Air Force Engineering University (AFEU), Xi'an, China, in 2013, and the Ph.D. degree in information and communication engineering from Tohoku University, Sendai, Japan, in 2019.

He is currently an Associate Professor with AFEU and the Leader of an Youth Innovation Team of Shaanxi Universities. His current research interests include radar target imaging, detection, and recognition.



Xiaowei Hu received the B.S. degree in radar engineering M.S. degree in electronic information, and Ph.D. degree in information and communication engineering from Air Force Engineering University, Xi'an, China, in 2010, 2012, and 2016, respectively.

Since 2017, he has been an Associate Professor with Air Force Engineering University. His research interests include radar imaging and target recognition.